

SARATOV TSENTRALNY, Russian Federation

WMO: 341720

Lat: 51.5659N

Lon: 46.0389E

Elev: 152

StdP: 99.51

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.3	-20.7	-26.4	0.3	-22.7	-23.8	0.4	-20.1	11.7	-5.8	10.4	-5.8	3.9	310	0.605

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	33.5	19.8	31.6	19.5	29.8	19.0	21.4	29.4	20.7	28.5	19.9	27.4	4.1	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.0	14.0	23.9	18.1	13.2	23.4	17.2	12.5	23.1	62.8	29.6	60.2	28.5	57.7	27.2	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.6	7.6	-25.8	35.8	3.6	2.1	-28.3	37.3	-30.4	38.5	-32.5	39.7	-35.1	41.2		
(5)			-26.1	23.4	3.6	1.0	-28.7	24.1	-30.8	24.7	-32.8	25.3	-35.4	26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-8.5	-8.2	-2.1	8.5	16.3	20.5	22.7	21.5	15.2	7.3	-0.3	-6.1		
	DBStd	12.52	6.39	6.45	5.13	5.37	4.87	4.33	3.62	4.22	4.53	4.58	5.23	6.07		
	HDD10.0	2482	573	509	375	91	7	1	0	0	8	110	309	498		
	HDD18.3	4428	832	742	633	297	99	27	6	18	116	343	559	757		
	CDD10.0	1499	0	0	0	46	201	314	393	356	163	25	0	0		
	CDD18.3	404	0	0	0	2	35	91	141	115	21	0	0	0		
	CDH23.3	3535	0	0	0	11	273	797	1276	1039	139	0	0	0		
	CDH26.7	1290	0	0	0	1	67	284	497	412	29	0	0	0		
(14) Wind	WSAvg	4.3	4.5	4.6	4.6	4.4	4.2	4.0	3.8	3.8	4.0	4.3	4.6	4.6		
(15) Precipitation	PrecAvg	431	39	29	27	24	37	46	41	42	38	33	42	39		
	PrecMax	629	86	75	78	51	90	142	125	118	113	88	105	116		
	PrecMin	255	3	1	1	4	1	9	8	0	1	0	5	8		
	PrecStd	94	19	19	17	14	25	33	26	28	28	23	24	19		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.6	3.2	12.8	25.0	30.9	35.1	36.0	36.1	29.8	20.6	12.1	5.9		
		MCWB	2.9	2.0	8.0	14.3	17.6	19.6	20.2	20.0	16.8	13.7	10.3	5.4		
	2%	DB	2.0	2.1	8.7	22.1	28.6	32.1	33.2	33.2	27.0	18.0	9.7	3.2		
		MCWB	1.4	1.2	5.4	12.6	17.2	19.6	20.1	19.8	16.4	12.2	8.3	2.5		
	5%	DB	1.0	1.2	6.1	19.2	26.4	29.9	31.2	31.0	24.8	15.8	7.9	2.1		
		MCWB	0.4	0.5	3.8	11.5	16.4	19.2	19.8	19.3	15.5	11.3	7.0	1.5		
	10%	DB	0.1	0.2	4.0	16.9	24.2	27.9	29.2	28.9	22.2	13.8	6.1	1.1		
		MCWB	-0.4	-0.3	2.2	10.1	15.4	18.5	19.2	18.6	14.6	10.3	5.3	0.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.0	2.4	8.1	15.6	19.7	22.7	22.5	22.0	19.2	15.6	10.8	5.2		
		MCDB	3.3	2.8	11.7	22.4	27.6	29.3	30.8	30.5	24.8	18.5	11.5	5.5		
	2%	WB	1.3	1.3	5.7	13.5	18.2	21.2	21.4	20.9	17.5	13.3	8.4	2.7		
		MCDB	1.6	1.8	7.9	20.0	26.3	28.5	29.8	29.3	23.3	16.1	9.1	3.1		
	5%	WB	0.3	0.6	4.0	12.2	17.2	20.2	20.7	20.1	16.4	11.9	6.9	1.6		
		MCDB	0.7	1.1	5.7	18.2	24.8	27.4	28.7	28.4	22.6	15.0	7.6	2.0		
	10%	WB	-0.4	-0.2	2.4	10.7	16.1	19.3	20.0	19.4	15.3	10.6	5.2	0.6		
		MCDB	0.1	0.3	3.7	15.9	22.7	26.2	27.3	27.0	21.3	13.2	5.8	1.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.0	5.5	5.9	8.7	9.5	9.5	9.7	9.9	8.6	6.6	4.6	4.7		
		MCDBR	4.4	4.5	7.8	11.9	11.5	11.8	12.0	12.3	11.5	9.5	5.8	4.1		
	5% WB	MCWBR	4.3	4.5	5.6	6.1	5.5	5.0	4.5	4.5	5.3	5.7	5.0	4.0		
		MCWBR	4.4	4.4	7.1	10.8	10.8	10.5	10.4	10.8	10.0	8.0	5.5	3.9		
(40) Clear-Sky Solar Irradiance	taub		0.299	0.331	0.365	0.429	0.400	0.387	0.404	0.422	0.376	0.350	0.327	0.301		
		taud	2.113	2.033	2.084	2.109	2.302	2.376	2.329	2.262	2.366	2.376	2.286	2.148		
	Ebn at Noon		678	758	813	804	854	868	845	805	806	752	655	607		
		Edh at Noon	77	111	128	142	123	115	119	121	97	79	66	64		
(44) All-Sky Solar Radiation	RadAvg		0.95	2.05	3.38	4.43	5.78	6.25	6.15	5.20	3.54	1.87	0.82	0.56		
	RadStd		0.21	0.33	0.56	0.39	0.51	0.53	0.48	0.40	0.45	0.28	0.14	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+186	+111
(47) Regional (5 neighbors)		+0.69	N/A	N/A	+1.25	N/A	N/A	N/A	N/A	+179	+77

Nomenclature: See separate page